

Dr. Émile Emery

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Nationalities: French, Swiss
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Education

- 2026** **French National Qualification for Associate Professor (CNU)**
Sections 28 (Condensed Matter and Materials) and 29 (Elementary Constituents)
- 2022–2025** **PhD in Theoretical and Numerical Physics**, defended on 8 October 2025
Université Paris-Saclay / CEA IRAMIS/SPEC, Gif-sur-Yvette
Statistical physics and resource investments in networks for the energy transition.
Supervisors: S. Aumaître and H. Bercegol. NNT: 2025UPASP078
- 2021–2022** **Master’s Year 2 — ICFP** (*Theoretical Physics programme*)
École Normale Supérieure, Paris
- 2020–2021** **Master’s Year 1 — High Energy Physics**
École Polytechnique, Palaiseau
- 2017–2020** **Magistère in Physics** (selective advanced undergraduate programme)
Université Paris Cité (*M1 interrupted in 2018–2019, see Other Training*)
- 2015–2017** **Preparatory classes MPSI-MP**, top-ranked in physics (MP)
Lycée Chaptal, Paris

Professional Experience

- Oct. 2025–present** **Postdoctoral Researcher**, Université Paris Cité
Paris Interdisciplinary Energy Research Institute (LIED/PIERI)
Supervisor: José Halloy. Funded by PEPR Recyclage. *Modelling of low and middle voltage networks at international scale.*
- Oct. 2022–Sept. 2025** **PhD Researcher**, CEA IRAMIS/SPEC, Gif-sur-Yvette
SPHYNX Group. Supervisors: Sébastien Aumaître, Hervé Bercegol.
Funded by CEA FOCUS TIME. *Statistical physics and resource investments in networks for the energy transition.*
- Apr.–Aug. 2022** **Research Intern**, Raman Research Institute (RRI), Bangalore, India
Quantum Gravity Group. Supervisor: Sumati Surya. *Analytical work on complex sequential growth in the framework of causal set theory (quantum gravity / graph theory).*
- Mar.–Jul. 2021** **Research Intern**, Institut d’Astrophysique de Paris (IAP)
Cosmology and Gravitation Group. Supervisor: Guillaume Faye. *Perturbative computations in Mathematica to estimate post-Newtonian spin effects on compact binary systems.*
- May–Jul. 2018** **Research Intern**, Astroparticles & Cosmology (APC), Paris
Cosmology Group. Supervisor: Cyrille Rosset. *Python-based image analysis for the Euclid project.*

Publications

PhD thesis

Emery É. *Statistical physics and resource investments in networks for the energy transition.* PhD thesis, Université Paris-Saclay, defended on 8 October 2025. NNT: 2025UPASP078.

Peer-reviewed journal articles

Emery É., Le Bihan J., Halloy J. *Connecting electrical grid length and material stock to population density: Comparison of a French-calibrated scaling with 35 electrified countries*. **Physical Review X Energy**, 2026. doi:10.1103/2xgs-8821

Emery É., Aumaître S., Bercegol H. *Power grid network growth model based on rural–urban differentiation*. **Physical Review X Energy** 4, no. 3, 033014, 2025. doi:10.1103/kysp-hgxt

Emery É., Bercegol H., Jonquères N., Aumaître S. *Complex network analysis of transmission networks preparing for the energy transition: application to the current French power grid*. **The European Physical Journal B**, vol. 97, no. 201, 2024. doi:10.1140/epjb/s10051-024-00837-7

Preprints and manuscripts under revision

Caligiuri A., **Emery É.** et al. *Time-varying ecological interactions characterise equilibrium and stability*. arXiv:2506.22123, 2025.

Azaiez A., **Emery É.** *Friend-of-friend generative models for social group interactions*. Under revision at **Physical Review E**. hal:05487223

Emery É., Aumaître S., Bercegol H. *Energy-optimal placement of storage systems in transmission grids*. Under revision at **Journal of Energy Storage**. arXiv:2603.03179

Manuscripts in preparation

Emery É., Le Bihan J., Halloy J. *Resource investment in deployment and renewal of worldwide long-term energy grids*.

Emery É., Presigny C., De Domenico M. *Universal generative modeling for inhomogeneous node distribution in spatial networks*.

Emery É., Halloy J. *Spatial scaling laws for multilevel distribution networks*.

Emery É., Jonquères N. *Network complexity of industrial metabolism*.

Emery É. *A large-scale open-access dataset of competitive social networks* (data paper).

Oral communications and posters

Jun. 2026 *Journées françaises des Systèmes Complexes*, Lyon. [Talk]

Feb. 2026 Invited seminar, CoMuNe Lab, University of Padua, Italy. [Invited talk]

Jan. 2026 *Winter Workshop on Complex Systems*, Mallorca, Spain. [Poster]

Jan. 2026 Invited seminar, LIED, Université Paris Cité. [Invited talk]

Jul. 2025 *Focus Time Seminar*, Grenoble. [Talk + Poster]

Jul. 2025 *10th Mediterranean School of Complex Networks*, Syracuse, Italy. [Talk]

Jun. 2025 *NetSci 2025 — International School and Conference on Network Science*, Maastricht, Netherlands. [Poster]

Apr. 2025 *4th MOMENTOM International Congress*, Université Paris-Saclay. [Talk]

Jul. 2024 *Focus Time Seminar*, Aix-les-Bains. [Talk + Poster]

May 2024 Invited seminar, *Networks & Games* research group, Centre d'Économie de la Sorbonne, Paris. [Invited talk]

Mar. 2024 Doctoral school “Énergie & Recherche”, Roscoff. [Talk]

Jan. 2024 *NetSciX 2024 — Intern. School and Conf. on Network Science*, Venice, Italy. [Talk]

Oct. 2023 *Focus Time Seminar*, Aix-les-Bains. [Talk]

Oct. 2023 *Journée Systèmes et Matières Complexes*, LPS, Orsay. [Talk]

Jul. 2023 *Exergy Economics Workshop*, Chambéry. [Talk]

Mar. 2023 *3rd MOMENTOM International Congress*, Université Paris-Saclay. [Poster]

Nov. 2022 *Young Physicists' Meeting*, Collège de France, Paris. [Talk]

Awards and Distinctions

- 2026 Admissibility for Associate Professor position in LIED/PIERI
- 2026 Selected participant, *MSCA MaRaThoN* (intensive 3-day MSCA proposal training), University of Padua
- 2023 Best Poster Award, *3rd MOMENTOM International Congress*, Université Paris-Saclay

Scientific and Academic Responsibilities

- Co-organiser of the *Winter Workshop on Complex Systems 2027*, Briançon
- Co-organiser of the *LIED Days 2026*, Université Paris Cité
- Co-organiser of a symposium for the *LIED Manifesto* project 2026, Université Paris Cité
- Reviewer: *Nature Communications Earth & Environment* (Nature Portfolio, 2026)
- Open science practices: code deposited on trusted repositories.

Teaching

Period	Volume	Course / Institution / Level
Mar.–Jun. 2025	13.5 h TD equiv.	Quantum and Statistical Physics, 1st-year engineering (BSc Year 3), CentraleSupélec
Dec. 2024	10 h TD equiv.	General Physics, BSc Mathematics & Computer Science, Université Paris Cité
Oct.–Dec. 2024	30.5 h TD equiv.	Optics, BSc Biomedical Sciences, Université Paris Cité
Mar.–Jun. 2024	13.5 h TD equiv.	Quantum and Statistical Physics, 1st-year engineering (BSc Year 3), CentraleSupélec
2015–2024	25–250 h/year	Private tutoring in mathematics and physics, secondary school to first-year bachelor's

Supervision

- Spring 2026 Co-supervision of a Master's internship at LIED (with P. Chatzimpiros, H. Bercegol and S. Aumaître): *Modelling the railway development during the industrial revolution*.

Science Outreach

- Dec. 2025 Outreach text produced for the *LIED Manifesto*.
- 5 Dec. 2025 Science discovery session designed for Year 2 primary school pupils.
- 20 Dec. 2024 *Art & Mathématiques* conference aimed at artists: visual introduction to network theory and the energy transition.
- 8 Oct. 2023 Introduction to chaotic processes, *Fête de la Science*, ENS Paris-Saclay.
- 2021–2022 Member of *Cours aux Ernest*: introductory physics lectures for ENS Paris students.
- 2021 Contribution to a science outreach article in the *Huffington Post*: *Le muon, particule révolutionnaire qui agite le monde de la physique*.
- 2019 Articles in the *Alma Mater* journal: *La théorie des mondes parallèles : de l'autre côté de l'horizon cosmologique* and *Un aperçu des abysses de l'univers*.

Technical Skills

Programming	Python, L ^A T _E X, Mathematica, Bash, Git, CAML, SQL — Complex networks and hypergraphs: <code>networkx</code> , <code>xgi</code> . — Large-scale data processing and visualisation: <code>pandas</code> , <code>geopandas</code> . — Machine learning and regression: <code>scikit-learn</code> , <code>tensorflow</code> .
Software	Visual Studio Code, Gephi, Logseq, Zotero, Activity Browser (life-cycle analysis), Anki, Pack Office, GitHub, Inkscape, Keybase.
Peer review	IOP Publishing Peer Review Excellence training.
Open science	Training in Data Management Plans and open science fundamentals.
Languages	French (mother tongue), English (fluent — publications and conferences), Spanish, Italian, Lojban (beginner).

Other Training

Philosophy	Registered auditor in the LOPHISS-SPH Master's programme (Philosophy and History of Science), Université Paris Diderot, 2018–2019.
Pedagogy	Training in the cognitive sciences of learning (MOOCs, doctoral school courses).
Other	Alto saxophone (conservatory and Arpej Jazz School, 2018–2019); PSC1 first-aid certificate; EU driving licence (category B).